

T0 Model: Complete Parameter-Free Particle Mass Calculation

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Direct Geometric Method vs. Extended Yukawa Method With Complete Neutrino Quantum Number Analysis and QFT Derivation

Abstract

The T0 model provides two mathematically equivalent but conceptually different calculation methods for particle masses: the direct geometric method and the extended Yukawa method. Both approaches use only $\xi = \frac{4}{3} \times 10^{-4}$ as geometric constant; the rational coefficients r_i are fixed from measured masses in the establishment phase. The documentation also contains an approach to the neutrino quantum numbers and a quantum field theoretical derivation of the ξ constant through EFT matching and 1-loop calculations. The systematic treatment of all particles, including neutrinos with the assumed double ξ suppression, shows the universal structure of the approach. The average deviation of less than 1% across the charged fermions means, compared with over twenty free Standard Model parameters, a clear reduction of the input quantities.

Note on versions. The German version of this document is more extensive than this English one. Earlier revisions shortened or changed parts of the English version. Where the two differ, the more extensive German version applies.

Status markers. Statements marked **[S]** are *speculative*: a hint or conjecture that has not yet been derived or numerically checked in the corpus.

Note on the calculation review (29 Sept. 2026). Individual statements:

- The neutrino values 9.1/1.9/18.0 meV are the same as in the neutrino part of Doc. 006, which R109 lists as not tenable; they follow from $1/\xi_\nu$ with no common anchor, their sum is 29.0, not 29.8 meV, and they are incompatible with the oscillation data (heaviest neutrino at least 50 meV). Authoritative is Doc. 340.
- A charged lepton of a 4th generation at 5.7 GeV is experimentally excluded (LEP, bound about 100 GeV); the calculation itself is correct.

The rest of the text is unchanged; see the correction register Doc. 190, R109, R134.

.1 Introduction

Particle physics faces a fundamental problem: the Standard Model with its over twenty free parameters offers no explanation for the observed particle masses. These appear arbitrary and without theoretical justification. The T0 model responds with two complementary calculation methods that work with the single geometric parameter ξ and also contain an approach to neutrino masses.

The Parameter Problem of the Standard Model

Despite its experimental success, the Standard Model has a theoretical gap: it contains more than 20 free parameters that must be determined experimentally. These include:

- **Fermion masses:** 9 charged lepton and quark masses
- **Neutrino masses:** 3 neutrino mass eigenvalues

- **Mixing parameters:** 4 CKM and 4 PMNS matrix elements
- **Gauge couplings:** 3 fundamental coupling constants
- **Higgs parameters:** Vacuum expectation value and self-coupling
- **QCD parameters:** Strong CP phase and others

Important

Parameter Reduction The T0 model traces the more than twenty free parameters of the Standard Model back to the single geometric parameter $\xi = \frac{4}{3} \times 10^{-4}$, which follows from the geometry of three-dimensional space. In addition there are rational coefficients r_i , which were fixed from the measured masses in the establishment phase (see the section on methodological clarification). This version also contains an approach to the neutrino quantum numbers as well as the quantum field theoretical derivation.

.2 Methodological Clarification: Establishment vs. Prediction

Important

Scientific-Historical Classification The T0 model is guided by the methodology of **pattern recognition and systematic classification**, analogous to the development of the periodic table (Mendeleev 1869) or the quark model (Gell-Mann 1964).

Two-Phase Development

Phase 1: Establishing the Systematics

1. Pattern recognition in known particle masses (electron, muon, tau)
2. Parameter determination from experimental data
3. Quantum number assignment establishment
4. Demonstration of mathematical equivalence of both methods

Phase 2: Unfolding Predictive Power

1. Extrapolation to unknown particles
2. Quark sector derivation from lepton patterns
3. New generation predictions
4. Experimental testing

Historical Precedent of Successful Pattern Physics

The T0 model is guided by the methodology of earlier pattern discoveries:

Discovery		Pattern Recognition	Predictions	Confirmation
Periodic Table (1869)		Atomic weights and properties	Gallium, Germanium, Scandium	Experimentally confirmed
Spectral Lines (1885)		Hydrogen lines	Rydberg formula for all series	Quantum mechanics
Quark Model (1964)		Hadron masses	Eightfold way	QCD theory
T0 Model (2025)		Lepton masses	4th generation, quarks	Experimental tests

Table 1: Historical precedent of pattern physics

.3 From Energy Fields to Particle Masses

The Fundamental Challenge

A core aim of the T0 model is to calculate particle masses from geometric principles. While the Standard Model requires over 20 free parameters to describe particle masses, the T0 model works with the geometric constant $\xi_{\text{geom}} = \frac{4}{3} \times 10^{-4}$ and rational coefficients r_i .

Mass Description

Parameter Reduction:

- **Standard Model:** 20+ free mass parameters (determined empirically)
- **T0 Model:** one geometric parameter ξ , plus rational coefficients r_i
- **Experimental Accuracy:** 99% average agreement (charged fermions; neutrinos compared with upper limits only)
- **Theoretical Foundation:** Three-dimensional space geometry + QFT derivation

Energy-Based Mass Concept

In the T0 framework, it is assumed that what we traditionally call “mass” is a manifestation of characteristic energy scales of field excitations:

$$m_i \rightarrow E_{\text{char},i} \quad (\text{characteristic energy of particle type } i) \quad (1)$$

This reinterpretation removes the distinction between mass and energy and treats both as different aspects of the same fundamental quantity.

.4 Two Complementary Calculation Methods

The T0 model provides two mathematically equivalent but conceptually different approaches to calculating particle masses:

Method 1: Direct Geometric Resonance

Conceptual Foundation: Particles as resonances in the universal energy field

The direct method treats particles as characteristic resonance modes of the energy field $E(x, t)$, analogous to standing wave patterns:

$$\text{Particles} = \text{Discrete resonance modes of } E(x, t)(x, t) \quad (2)$$

Three-Step Calculation Process:

Step 1: Geometric Quantization

$$\xi_i = \xi_0 \cdot f(n_i, l_i, j_i) \quad (3)$$

where:

$$\xi_0 = \frac{4}{3} \times 10^{-4} \quad (\text{base geometric parameter}) \quad (4)$$

$$n_i, l_i, j_i = \text{quantum numbers from 3D wave equation} \quad (5)$$

$$f(n_i, l_i, j_i) = \text{geometric function from spatial harmonics} \quad (6)$$

Step 2: Resonance Frequencies

$$\omega_i = \frac{c^2}{\xi_i \cdot r_{\text{char}}} \quad (7)$$

In natural units ($c = 1$):

$$\omega_i = \frac{1}{\xi_i} \quad (8)$$

Step 3: Mass Determination from Energy Conservation

$$E_{\text{char},i} = \hbar \omega_i = \frac{\hbar}{\xi_i} \quad (9)$$

In natural units ($\hbar = 1$):

$$E_{\text{char},i} = \frac{1}{\xi_i} \quad (10)$$

Method 2: Extended Yukawa Method

Conceptual Foundation: Bridge to Standard Model formulation

The extended Yukawa method maintains compatibility with Standard Model calculations while tracing the Yukawa couplings back to a geometric structure:

$$E_{\text{char},i} = y_i \cdot v \quad (11)$$

where $v = 246$ GeV is the Higgs vacuum expectation value.

Geometric Yukawa Couplings:

$$y_i = r_i \cdot \left(\frac{4}{3} \times 10^{-4} \right)^{\pi_i} \quad (12)$$

Generation Hierarchy:

$$\text{1st Generation: } \pi_i = \frac{3}{2} \quad (\text{electron, up quark}) \quad (13)$$

$$\text{2nd Generation: } \pi_i = 1 \quad (\text{muon, charm quark}) \quad (14)$$

$$\text{3rd Generation: } \pi_i = \frac{2}{3} \quad (\text{tau, top quark}) \quad (15)$$

The coefficients r_i are simple rational numbers assigned to the geometric structure of each particle type.

.5 Quantum Field Theoretical Derivation of the ξ Constant

EFT Matching and Yukawa Coupling after EWSB

After electroweak symmetry breaking we have the Yukawa interaction:

$$\mathcal{L}_{\text{Yukawa}} \supset -\lambda_h \bar{\psi} \psi H, \quad \text{with} \quad H = \frac{v+h}{\sqrt{2}} \quad (16)$$

After EWSB:

$$\mathcal{L} \supset -m \bar{\psi} \psi - y h \bar{\psi} \psi \quad (17)$$

with the relations:

$$m = \frac{\lambda_h v}{\sqrt{2}} \quad \text{and} \quad y = \frac{\lambda_h}{\sqrt{2}} \quad (18)$$

The local mass dependence on the physical Higgs field $h(x)$ leads to:

$$m(h) = m \left(1 + \frac{h}{v} \right) \Rightarrow \partial_\mu m = \frac{m}{v} \partial_\mu h \quad (19)$$

T0 Operators in Effective Field Theory

In T0 theory, operators of the form appear:

$$O_T = \bar{\psi} \gamma^\mu \Gamma_\mu^{(T)} \psi \quad (20)$$

with the characteristic time field coupling term:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} \quad (21)$$

Inserting the Higgs dependence:

$$\Gamma_\mu^{(T)} = \frac{\partial_\mu m}{m^2} = \frac{1}{mv} \partial_\mu h \quad (22)$$

This suggests that a $\partial_\mu h$ -coupled vector current is the UV origin.

1-Loop Matching Calculation

The 1-loop amplitude for the T0 vertex yields:

$$F_V(0) = \frac{y^2}{16\pi^2} \left[\frac{1}{2} - \frac{1}{2} \ln \left(\frac{m_h^2}{\mu^2} \right) + r(r - \ln r - 1)/(r - 1)^2 \right] \quad (23)$$

For hierarchical masses ($m \ll m_h$) the constant term dominates:

$$F_V(0) \approx \frac{y^2}{32\pi^2} \quad (24)$$

Final ξ Formula from Higgs Physics

The EFT matching provides the fundamental relation:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \quad (25)$$

With standard Higgs parameters ($m_h = 125.1$ GeV, $v = 246.22$ GeV, $\lambda_h \approx 0.13$):

$$\xi \approx 1.318 \times 10^{-4} \quad (26)$$

This agrees well with the geometric determination $\xi_0 = \frac{4}{3} \times 10^{-4} \approx 1.333 \times 10^{-4}$ (deviation $\approx 1.15\%$).

.6 Universal Particle Mass Systematics

Revised Universal Fermion Table

Fermion	Generation	Family	Spin	r_f	Exponent p_f	Symmetry
Electron Neutrino	1	0	1/2	4/3	5/2	Double ξ
Electron	1	0	1/2	4/3	3/2	Lepton number
Muon Neutrino	2	1	1/2	16/5	3	Double ξ
Muon	2	1	1/2	16/5	1	Lepton number
Tau Neutrino	3	2	1/2	25/9	25/9	Double ξ
Tau	3	2	1/2	25/9	2/3	Lepton number
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	$\frac{25}{2}$	3/2	Color + Isospin
Charm	2	1	1/2	2*	2/3	Color
Strange	2	1	1/2	$\frac{26}{9}$	1	Color
Top	3	2	1/2	$\frac{1}{28}$	-1/3	Color
Bottom	3	2	1/2	$\frac{3}{2}$	1/2	Color

.7 Complete Numerical Reconstruction

The following analysis shows the explicit calculation of all fermions with both methods:

Foundations and Experimental Input Data

Fundamental Constants:

$$\xi_0 = \xi = \frac{4}{3} \times 10^{-4} = 1.333333333... \times 10^{-4} \quad (27)$$

$$v = 246 \text{ GeV} \quad (28)$$

Experimental Masses (PDG-close values):

$$m_e^{\text{exp}} = 0.0005109989461 \text{ GeV} \quad (29)$$

$$m_\mu^{\text{exp}} = 0.1056583745 \text{ GeV} \quad (30)$$

$$m_\tau^{\text{exp}} = 1.77686 \text{ GeV} \quad (31)$$

^{0*} Corrected from originally 8/9 based on detailed numerical analysis

Charged Leptons: Detailed Calculations

Electron Mass Calculation:

Direct Method:

$$\xi_e = \frac{4}{3} \times 10^{-4} \times f_e(1, 0, 1/2) \quad (32)$$

$$= \frac{4}{3} \times 10^{-4} \times 1 = \frac{4}{3} \times 10^{-4} \quad (33)$$

$$E_e = \frac{1}{\xi_e} = \frac{3}{4 \times 10^{-4}} = 0.511 \text{ MeV} \quad (34)$$

Extended Yukawa Method:

$$r_e = \frac{m_e^{\text{exp}}}{v \cdot \xi^{3/2}} \approx 1.349 \quad (35)$$

$$y_e = 1.349 \times \left(\frac{4}{3} \times 10^{-4}\right)^{3/2} \quad (36)$$

$$E_e = y_e \times 246 \text{ GeV} = 0.511 \text{ MeV} \quad (37)$$

Muon Mass Calculation:

Direct Method:

$$\xi_\mu = \frac{4}{3} \times 10^{-4} \times f_\mu(2, 1, 1/2) \quad (38)$$

$$= \frac{4}{3} \times 10^{-4} \times \frac{16}{5} = \frac{64}{15} \times 10^{-4} \quad (39)$$

$$E_\mu = \frac{1}{\xi_\mu} = 105.66 \text{ MeV} \quad (40)$$

Extended Yukawa Method:

$$y_\mu = \frac{16}{5} \times \left(\frac{4}{3} \times 10^{-4}\right)^1 = 4.267 \times 10^{-4} \quad (41)$$

$$E_\mu = y_\mu \times 246 \text{ GeV} = 104.96 \text{ MeV} \quad (42)$$

Experiment: 105.66 MeV → Deviation $\approx 0.65\%$

Complete Neutrino Treatment

Neutrino Approach **[S]** The T0 model contains a geometric approach to neutrino masses based on the assumption of a characteristic **double ξ suppression**. This addresses the previous gap in the neutrino sector; the comparison with oscillation data is still outstanding.

Neutrino Quantum Numbers

Neutrinos follow the same quantum number structure as other fermions, but with an essential modification due to their weak interaction nature:

Neutrino	n	l	j	Suppression
ν_e	1	0	1/2	Double ξ
ν_μ	2	1	1/2	Double ξ
ν_τ	3	2	1/2	Double ξ

Table 2: Neutrino quantum numbers with characteristic double ξ suppression

Double ξ Suppression Mechanism

The central assumption is that neutrinos experience an additional geometric suppression factor:

$$f(n_{\nu_i}, l_{\nu_i}, j_{\nu_i}) = f(n_i, l_i, j_i)_{\text{Lepton}} \times \xi \quad (43)$$

Neutrino Mass Calculations:

Electron Neutrino:

$$\xi_{\nu_e} = \frac{4}{3} \times 10^{-4} \times 1 \times \frac{4}{3} \times 10^{-4} = \frac{16}{9} \times 10^{-8} \quad (44)$$

$$E_{\nu_e} = \frac{1}{\xi_{\nu_e}} = 9.1 \text{ meV} \quad (45)$$

Muon Neutrino:

$$\xi_{\nu_\mu} = \frac{4}{3} \times 10^{-4} \times \frac{16}{5} \times \frac{4}{3} \times 10^{-4} = \frac{256}{45} \times 10^{-8} \quad (46)$$

$$E_{\nu_\mu} = \frac{1}{\xi_{\nu_\mu}} = 1.9 \text{ meV} \quad (47)$$

Tau Neutrino:

$$\xi_{\nu_\tau} = \frac{4}{3} \times 10^{-4} \times \frac{25}{9} \times \frac{4}{3} \times 10^{-4} = \frac{400}{81} \times 10^{-8} \quad (48)$$

$$E_{\nu_\tau} = \frac{1}{\xi_{\nu_\tau}} = 18.0 \text{ meV} \quad (49)$$

.8 Complete Quark Analysis with Both Methods

Explicit Quark Mass Calculations

We use $\xi = \frac{4}{3} \times 10^{-4}$ and $v = 246 \text{ GeV}$. For the Yukawa representation:

$$y_i = r_i \xi p_i, \quad m_i^{\text{pred}} = y_i v.$$

For the direct geometric representation:

$$f_i = \frac{1}{\xi m_i^{\text{exp}}}, \quad m_i^{\text{exp}} = \frac{1}{\xi f_i}.$$

Quark	p_i	r_i (corr.)	m_i^{pred} (GeV)	m_i^{exp} (GeV)	rel. error (%)	Remark
Up	3/2	6	2.272×10^{-3}	2.27×10^{-3}	+0.11	OK
Down	3/2	25/2	4.734×10^{-3}	4.72×10^{-3}	+0.30	OK
Strange	1	26/9	9.50×10^{-2}	9.50×10^{-2}	0.00	Exact
Charm	2/3	2	1.279×10^0	1.28	-0.08	Corrected
Bottom	1/2	3/2	4.261×10^0	4.26	+0.02	OK
Top	-1/3	1/28	1.7198×10^2	171	+0.57	OK

Table 3: Yukawa predictions with corrected r_i, p_i and comparison with reference masses.

Charm Quark Correction

The originally tabulated value $r_c = 8/9$ does not reproduce the referenced mass $m_c = 1.28$ GeV. The required value is:

$$r_c^{\text{required}} = \frac{m_c^{\text{exp}}}{v\xi^{2/3}} \approx 1.994 \approx 2.$$

Therefore, $r_c \approx 2$ was inserted in the corrected universal table.

.9 Comprehensive Experimental Validation

Complete Accuracy Analysis

The following table compares the T0 values with the experimental values:

Particle	T0 Prediction	Experiment	Accuracy	Type
<i>Charged Leptons</i>				
Electron	0.511 MeV	0.511 MeV	99.98%	Lepton
Muon	104.96 MeV	105.66 MeV	99.35%	Lepton
Tau	1777.1 MeV	1776.86 MeV	99.99%	Lepton
<i>Neutrinos</i>				
ν_e	9.1 meV	< 450 meV	Compatible	Neutrino
ν_μ	1.9 meV	< 180 keV	Compatible	Neutrino
ν_τ	18.0 meV	< 18 MeV	Compatible	Neutrino
<i>Quarks</i>				
Up Quark	2.272 MeV	2.27 MeV	99.89%	Quark
Down Quark	4.734 MeV	4.72 MeV	99.70%	Quark
Strange Quark	95.0 MeV	95.0 MeV	100.0%	Quark
Charm Quark	1.279 GeV	1.28 GeV	99.92%	Quark
Bottom Quark	4.261 GeV	4.26 GeV	99.98%	Quark
Top Quark	171.99 GeV	171 GeV	99.43%	Quark
Average			99.6%	All Fermions

Table 4: Complete experimental validation of T0 model predictions

Key Result

Summary of Accuracy The T0 model achieves 99.6% average accuracy across the **charged** fermions with the single geometric parameter ξ and the tabulated rational coefficients r_i . The neutrino sector is included as an approach **[S]**; so far it is compared with upper limits only.

.10 Experimental Predictions and Precision Tests

Modified QED Vertex Corrections

The T0 theory predicts modified Feynman rules:

$$\text{Time field vertex: } -i\gamma^\mu T_\mu^{(T)} = i\gamma^\mu \frac{\partial_\mu m}{m^2} \quad (50)$$

$$\text{Modified fermion propagator: } S_F^{(T0)}(p) = S_F(p) \cdot \left[1 + \frac{\beta}{p^2} \right] \quad (51)$$

Neutrino Validation

The T0 neutrino predictions satisfy the limits listed in the table; a comparison with the measured mass differences from oscillation experiments is still outstanding in this document [S]:

Parameter	T0 Prediction	Experimental Limit	Status
m_{ν_e}	9.1 meV	< 450 meV (KATRIN)	Fulfilled
m_{ν_μ}	1.9 meV	< 180 keV (indirect)	Fulfilled
m_{ν_τ}	18.0 meV	< 18 MeV (indirect)	Fulfilled
$\sum m_\nu$	29.8 meV	< 60 meV (Cosmology 2024)	Fulfilled

Table 5: T0 neutrino predictions vs. experimental constraints

Important

Neutrino Mass Hierarchy [S] The T0 model orders the masses as $m_{\nu_\mu} < m_{\nu_e} < m_{\nu_\tau}$. How this assignment to flavour states relates to the **normal ordering** of the mass eigenstates preferred by oscillation data remains to be clarified.

.11 Predictive Power of the Established System

New Particle Generations

With established patterns, new particles can be predicted:

4th Generation (extrapolated) [S]:

$$n = 4, \quad \pi_4 = \frac{1}{2}, \quad r_4 \approx 2.0 \quad (52)$$

$$m_{4\text{th Gen}} = r_4 \times \xi^{1/2} \times v \approx 5.7 \text{ GeV} \quad (53)$$

Quark Sector Extrapolation

Lepton patterns can be transferred to quarks:

Quark	Generation	r_i	π_i	Prediction
Up	1	6	3/2	2.3 MeV
Down	1	12.5	3/2	4.7 MeV
Charm	2	2.0	2/3	1.3 GeV
Strange	2	2.89	1	95 MeV
Top	3	0.036	-1/3	173 GeV
Bottom	3	1.5	1/2	4.3 GeV

Table 6: Quark predictions from established patterns

.12 Corrected Interpretation of Mathematical Equivalence

Meaning of Equivalence The mathematical equivalence of both methods is **given by definition** when parameters (r_i or f_i) are determined from the same experimental masses. The equivalence is not proof of the theory, but a consistency property of the mathematical structure.

Transformation Relationship as Bridge

The fundamental relation:

$$f_i = \frac{1}{r_i \xi^{\pi_i} v \xi_0} \quad (54)$$

mathematically connects both methods. When r_i is determined from experimental masses, f_i follows automatically and vice versa.

Particle	m^{exp} (GeV)	r_i (Yukawa)	f_i (direct)	Accuracy
Electron	0.000511	1.349	1.468×10^7	99.98%
Muon	0.10566	3.221	7.099×10^4	99.35%
Tau	1.77686	2.778	4.221×10^3	99.99%
ν_e	9.1×10^{-6}	1.349	8.235×10^{10}	Prediction
ν_μ	1.9×10^{-6}	3.221	3.947×10^{11}	Prediction
ν_τ	18.0×10^{-6}	2.778	3.989×10^{10}	Prediction

Table 7: Numerical equivalence of both T0 methods for all leptons

.13 Methodological Foundation

Reversibility of the Established System

After the establishment phase, the T0 system is meant to become predictive:

Established Lepton Patterns:

$$\text{1st Generation (n=1): } \pi_i = \frac{3}{2}, \quad r_e \approx 1.35 \quad (55)$$

$$\text{2nd Generation (n=2): } \pi_i = 1, \quad r_\mu \approx 3.2 \quad (56)$$

$$\text{3rd Generation (n=3): } \pi_i = \frac{2}{3}, \quad r_\tau \approx 2.8 \quad (57)$$

Experimental Testability

T0 predictions are experimentally falsifiable:

1. **LHC searches:** New particles at characteristic energies (5-6 GeV range)
 2. **Precision measurements:** Refinement of r_i parameters
 3. **Neutrino tests:** Direct neutrino mass measurements
 4. **Anomalous magnetic moments:** T0 corrections to g-2 experiments
- Arguments for the T0 procedure:
1. **Systematic structure:** All parameters follow recognizable patterns
 2. **Predictive power:** After establishment, new particles become predictable
 3. **Experimental testability:** Predictions are falsifiable
 4. **QFT foundation:** Quantum field theoretical derivation of ξ constant
 5. **Historical precedent:** Methodology of pattern physics (periodic table, quark model)

.14 Parameter Structure and Universal Systematics

Important

Role of the Parameter ξ The parameter ξ can additionally be estimated from Higgs parameters:

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.318 \times 10^{-4} \quad (58)$$

Thus ξ cannot be chosen freely. The coefficients r_i are rational numbers; deriving them from geometry without recourse to measured masses is the next step.

Universal Quantum Number Table

Particle	n	l	j	r_i	p_i	Special
<i>Charged Leptons</i>						
Electron	1	0	1/2	4/3	3/2	–
Muon	2	1	1/2	16/5	1	–
Tau	3	2	1/2	25/9	2/3	–
<i>Neutrinos</i>						
ν_e	1	0	1/2	4/3	5/2	Double ξ
ν_μ	2	1	1/2	16/5	3	Double ξ
ν_τ	3	2	1/2	25/9	25/9	Double ξ
<i>Quarks</i>						
Up	1	0	1/2	6	3/2	Color
Down	1	0	1/2	25/2	3/2	Color + Isospin
Charm	2	1	1/2	2	2/3	Color
Strange	2	1	1/2	26/9	1	Color
Top	3	2	1/2	1/28	-1/3	Color
Bottom	3	2	1/2	3/2	1/2	Color

Table 8: Complete universal quantum number table for all fermions

.15 Critical Assessment and Limitations

Theoretical Open Questions

1. **Number of generations:** Why exactly three generations plus fourth prediction?
2. **Hierarchy problem:** Connection between different energy scales
3. **CP violation:** Incorporation of CKM and PMNS mixing matrices